

Bipartisan Infrastructure Law helping Alaska map critical mineral resources

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FULL TEXT

The funding comes from a \$74 million investment allocated earlier this year to the **USGS Mineral Resources Program’ s Earth Mapping Resources Initiative** (Earth MRI), provided through annual appropriations and investments from President Biden’ s Bipartisan Infrastructure Law. Overall, the Infrastructure Law is providing a \$510.7 million investment to USGS to advance scientific innovation and map critical minerals.

Funding announced today includes grants to the Alaska Division of Geological & Geophysical Surveys for geologic mapping and geochemical analyses for an area of the Yukon-Tanana Upland near the Canadian border. In addition, the USGS funding will support new airborne geophysical surveys in the Kuskokwim River region and in the Seward Peninsula. This data will help improve our understanding of the area’ s critical mineral resources, a key step in securing a reliable and sustainable supply of the critical minerals that power everything from household appliances and electronics to clean energy technologies like batteries and wind turbines.

“Resources from President Biden’ s Bipartisan Infrastructure Law are helping to advance scientific innovation and help us secure a reliable and sustainable supply of critical minerals,” said **USGS director David Applegate**. “All the way back to the days of the Gold Rush, Alaska has been famous for its mineral wealth. These new projects represent the next steps in understanding the mineral potential for commodities that are critical to our national economy and defense. Data provided through these projects will have many applications and will create a foundation for better understanding of mineral and geothermal resources, earthquake hazard potential, carbon storage capacity and many other geoscience opportunities.”

The Alaska Division of Geological & Geophysical Surveys will focus on geologic mapping for critical mineral commodities. The geologic mapping efforts, which are managed through the **USGS National Cooperative Geologic Mapping Program**, will refine our understanding of the geology underlying areas of interest. In addition to helping identify mineral potential, these maps also support decisions about land use, provide information on water, energy and minerals resources and can help mitigate the impact of geologic hazards on communities.

The Alaska Division of Geological & Geophysical Surveys and the USGS will also conduct airborne magnetic and radiometric surveys of the Kuskokwim River region, which has potential for antimony, gold, rare earth elements, tin, tungsten, and other critical minerals, and an airborne electromagnetic survey in the Seward Peninsula in areas with potential for graphite.

Airborne geophysical surveys flown as part of Earth MRI collect a combination of magnetic, radiometric, and electromagnetic data. Magnetic data can tell us the amount of magnetic minerals, primarily magnetite, in exposed and deeply buried rocks, whereas the radiometric surveys detect differential levels of natural radioactivity in exposed rocks that can tell us the relative amounts of potassium, uranium and thorium present. Airborne electromagnetic surveys help identify areas that are electrically conductive, such as areas with graphite in the near surface. This information allows scientists to help identify locations of particular rocks that may be likely to host minerals of interest, as well as locate geothermal energy resources, groundwater resources, and potential earthquake hazards in the region.

The critical mineral commodities that are included in the research projects in Alaska are:

- **Arsenic:** used in lumber preservatives, pesticides, and semi-conductors

- **Antimony:** used in flame-proofing compounds, alloys, and batteries
- **Bismuth:** used in medical and atomic research
- **Cobalt:** used in rechargeable batteries and superalloys
- **Graphite:** used for lubricants, batteries, and fuel cells
- **Indium:** mostly used in LCD screens
- **Platinum group metals:** used for catalytic agents
- **Rare earth elements:** primarily used in batteries and electronics
- **Tantalum:** used in electronic components, mostly capacitors
- **Tellurium:** used in steelmaking and solar cells
- **Tin:** used as protective coatings and alloys for steel
- **Tungsten:** primarily used to make wear-resistant metals

Earth MRI is a partnership between the USGS and state geological surveys across America to modernize our understanding of the Nation's fundamental geologic framework and mineral resources through new geologic maps, geophysical and topographic surveys, and geochemical sampling. The 2022 Bipartisan Infrastructure Law provided additional funding that has accelerated this new mapping in areas with potential for hosting critical mineral resources both still in the ground and in mine wastes.

More information can be found [here](#). To learn more about how the USGS is investing the resources from the Bipartisan Infrastructure Law, visit [our website](#). To learn more about USGS mineral resource and commodity information, please visit our [website](#) and follow us on [Twitter](#).

DETAILS

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